



HQ-MTV PROMPT STUDIO

Project report



JULY 31, 2026

Teachers

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REPORT

KPITB · AI & Machine Learning Training Program

Group Project Report

*Regression on Tabular Data ·
Pipeline*

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Track (Classification / Regression)	Regression
Project title	Medical Insurance Charges Prediction
Dataset name and source	Medical Cost Personal Insurance dataset (Kaggle — insurance.csv)
Colab notebook link (view access on)	https://colab.research.google.com/drive/1kmBp9m2siEZ4-2R_mVC62NXIgEZmc23s
Submission date	31 JULY 2026

2. Problem Statement

This project predicts an individual's annual medical insurance charges using six input features: age, sex, body mass index (BMI), number of children, smoking status, and region. Since the target variable, **insurance charges**, is a continuous numerical value rather than a category, this is a **regression** problem. The developed model estimates future healthcare costs based on personal and demographic characteristics. In real-world applications, insurance companies and healthcare providers can use these predictions to assess financial risk, determine appropriate premium rates, and support data-driven pricing decisions for new policyholders.

3. Summary

The dataset is the Medical Cost Personal Insurance dataset from Kaggle, with [1,338 records](#) and [7 columns](#); the target is charges, the individual's annual medical insurance cost, alongside six features covering demographics, health, and location. EDA showed charges is strongly right-skewed, with almost no missing data and only one duplicate row, and it is most closely tied to smoking status (correlation 0.83) and age (correlation 0.53), so charges was log-transformed before modelling to make the relationship easier for the models to learn. We compared a Ridge Regression baseline against a tuned Random Forest Regressor; Random Forest won clearly, cutting test-set RMSE from about \$8,681 to about \$4,598 and raising R2 from 0.51 to 0.86. Our headline metric is $R^2 = 0.86$ on the test set for the Random Forest model, meaning it explains roughly 86% of the variance in individual insurance charges.

(142 words)

4. Data Collection

The dataset used in this project is the **Medical Cost Personal Insurance Dataset**, obtained from **Kaggle**: <https://www.kaggle.com/datasets/mirichoi0218/insurance>. It contains **1,338 rows** and **7 columns**, including six input features and one target variable. The target column is **charges**, which represents an individual's annual medical insurance cost. The input features are **age**, **sex**, **BMI**, **number of children**, **smoking status**, and **region**. The dataset was loaded into Google Colab using the pandas library with `pd.read_csv()`. Initial exploration was performed using `df.shape` to verify its dimensions, `df.info()` to inspect data types and missing values, and `df.head()` to preview the first five records. Confirmed there were zero missing values in any of the seven columns, so no imputation was required at this stage.

5. Data Preprocessing &

Exploratory Data Analysis

Missing Values and Duplicates: The dataset was first examined for data quality using `df.isnull().sum()` and `df.duplicated().sum()`. No missing values were found across any of the seven columns, ensuring that no imputation was required. One duplicate record was identified and removed to eliminate redundancy and improve data consistency.

Encoding:

The categorical features—such as **sex**, **smoker**, and **region**—were transformed using **OneHotEncoder**. One-hot encoding was selected because these variables are nominal categories with no inherent order, allowing the models to process them without introducing misleading numerical relationships.

Scaling:

Numerical features were standardized using **StandardScaler**. Feature scaling was necessary for the **Ridge Regression** model because it is sensitive to differences in feature magnitudes. Although **Random Forest Regressor** does not require scaling due to its tree-based structure, the same preprocessing pipeline was applied for consistency across both models.

Splitting and Leakage:

The dataset was divided into training and testing sets using `train_test_split(test_size=0.2, random_state=42)` without stratification, as the target variable is continuous. To prevent data leakage, every preprocessing step, including encoding and scaling, was fitted only on the training data and then applied to the test data.

Exploratory Data Analysis:

The histogram revealed a right-skewed distribution of insurance charges, indicating the presence of high-cost outliers. The boxplot showed that smokers incur substantially higher medical charges than non-smokers. The correlation heatmap identified **smoking status** and **age** as the strongest predictors of insurance charges, while the scatter plot demonstrated that medical costs generally increase with age, particularly among smokers.

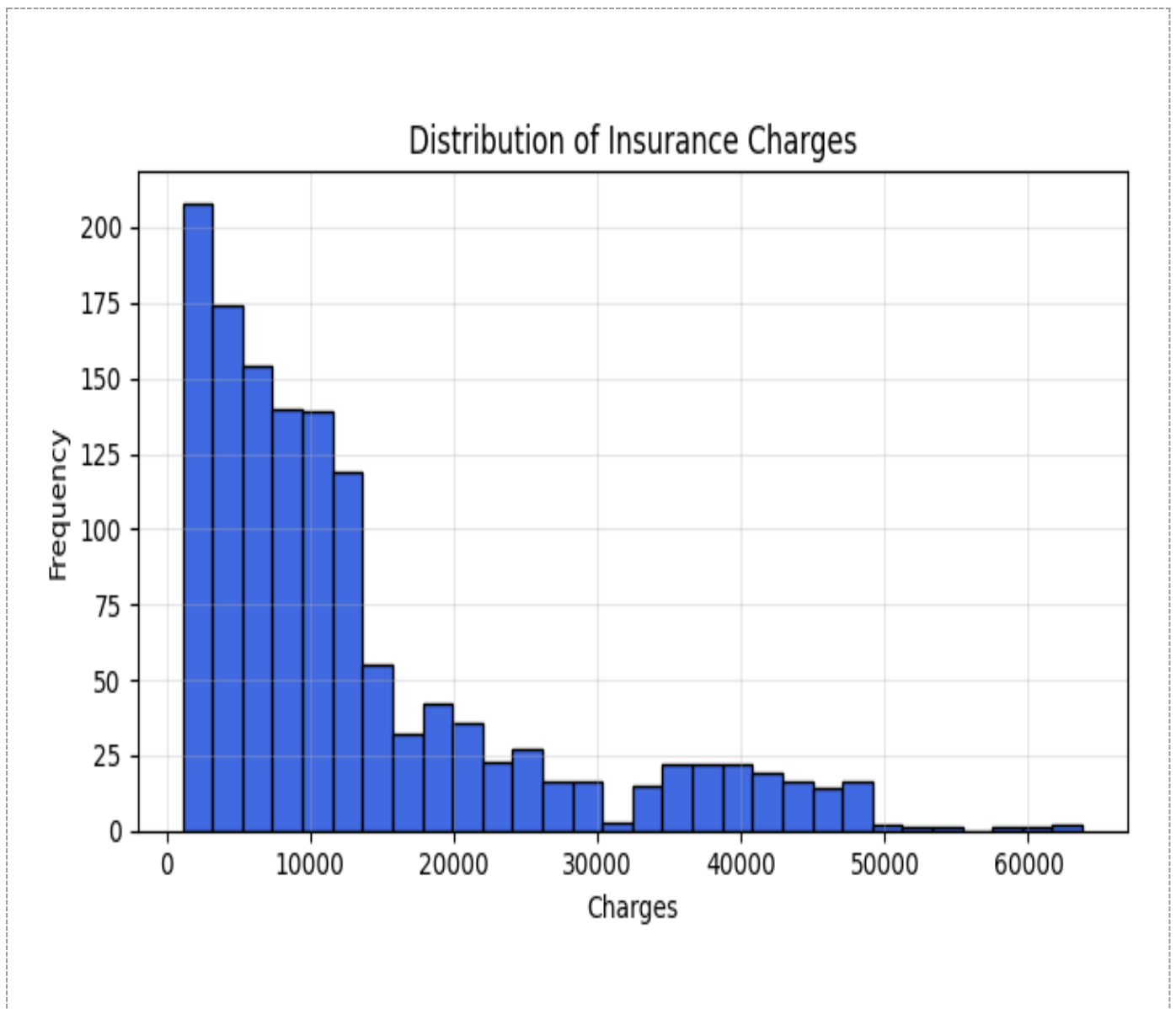


Figure 1 —

Charges is heavily right-skewed — most policyholders pay under \$15,000, but a long tail pays much more, which is why we modelled $\log(\text{charges})$.

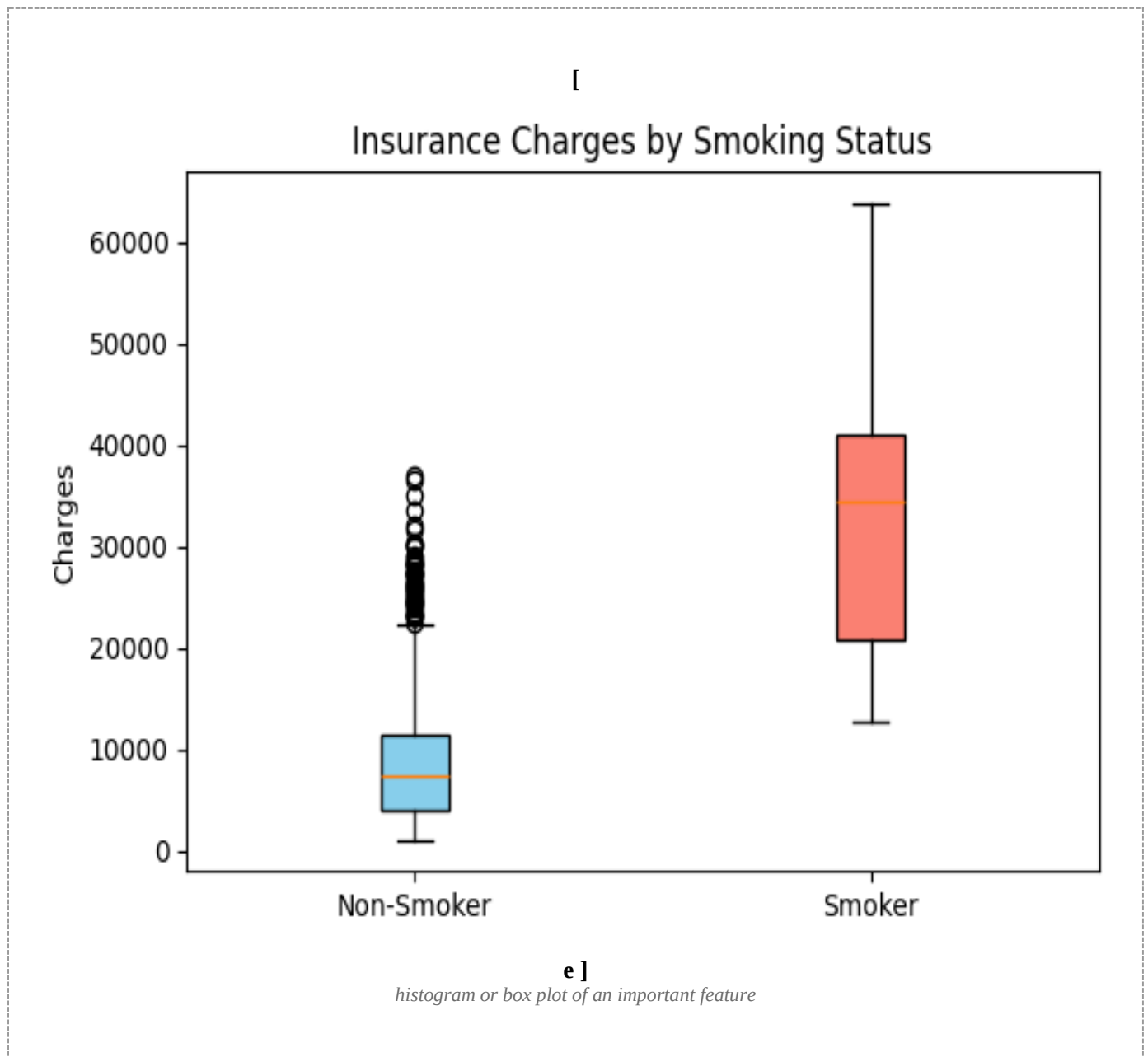


Figure 2

— Boxplot of charges by smoker status shows smokers' charges are far higher with almost no overlap with non-smokers.

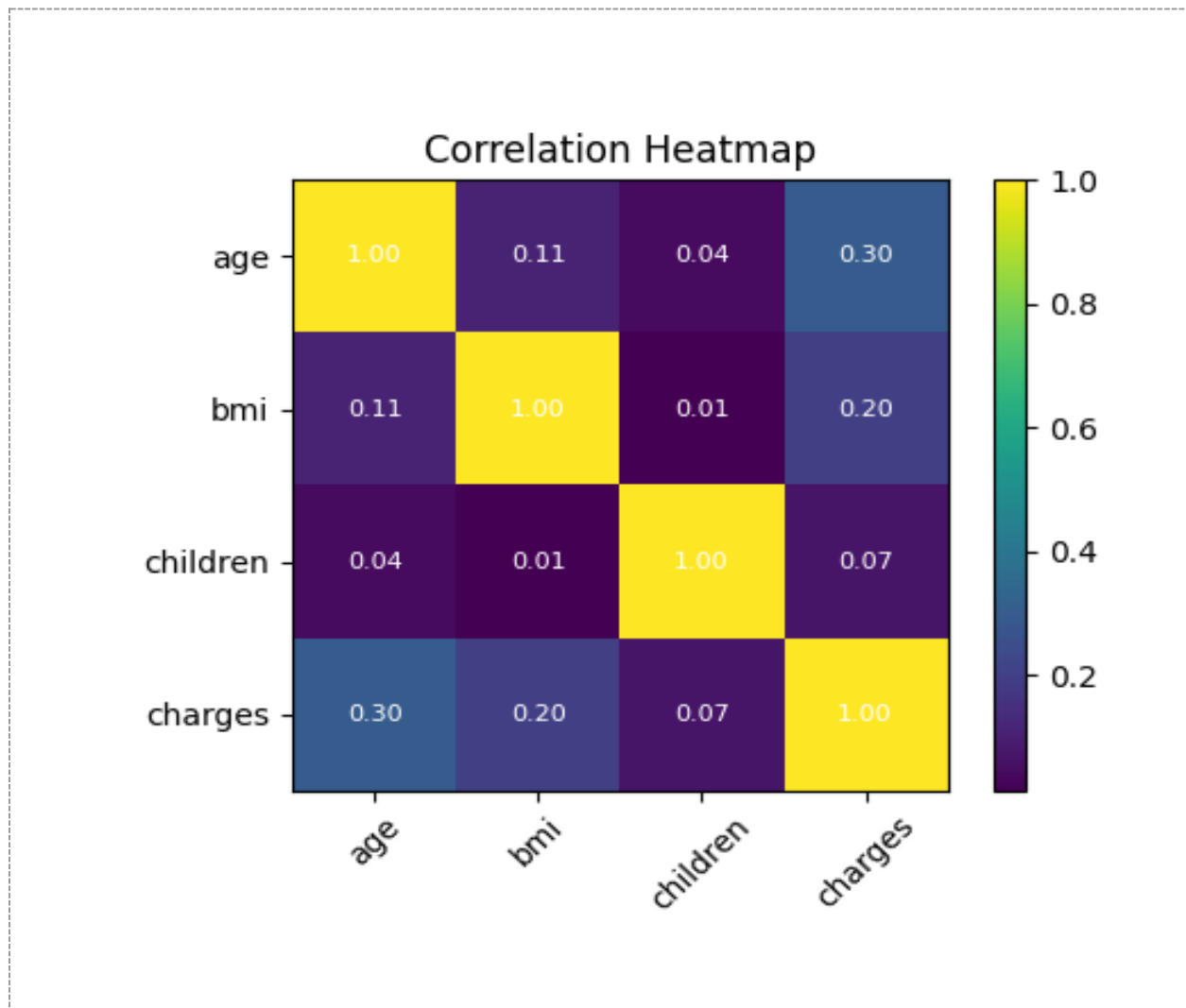


Figure 3 —

The heatmap confirms smoker (0.83) and age (0.53) are the strongest predictors of charges; bmi and children correlate weakly.

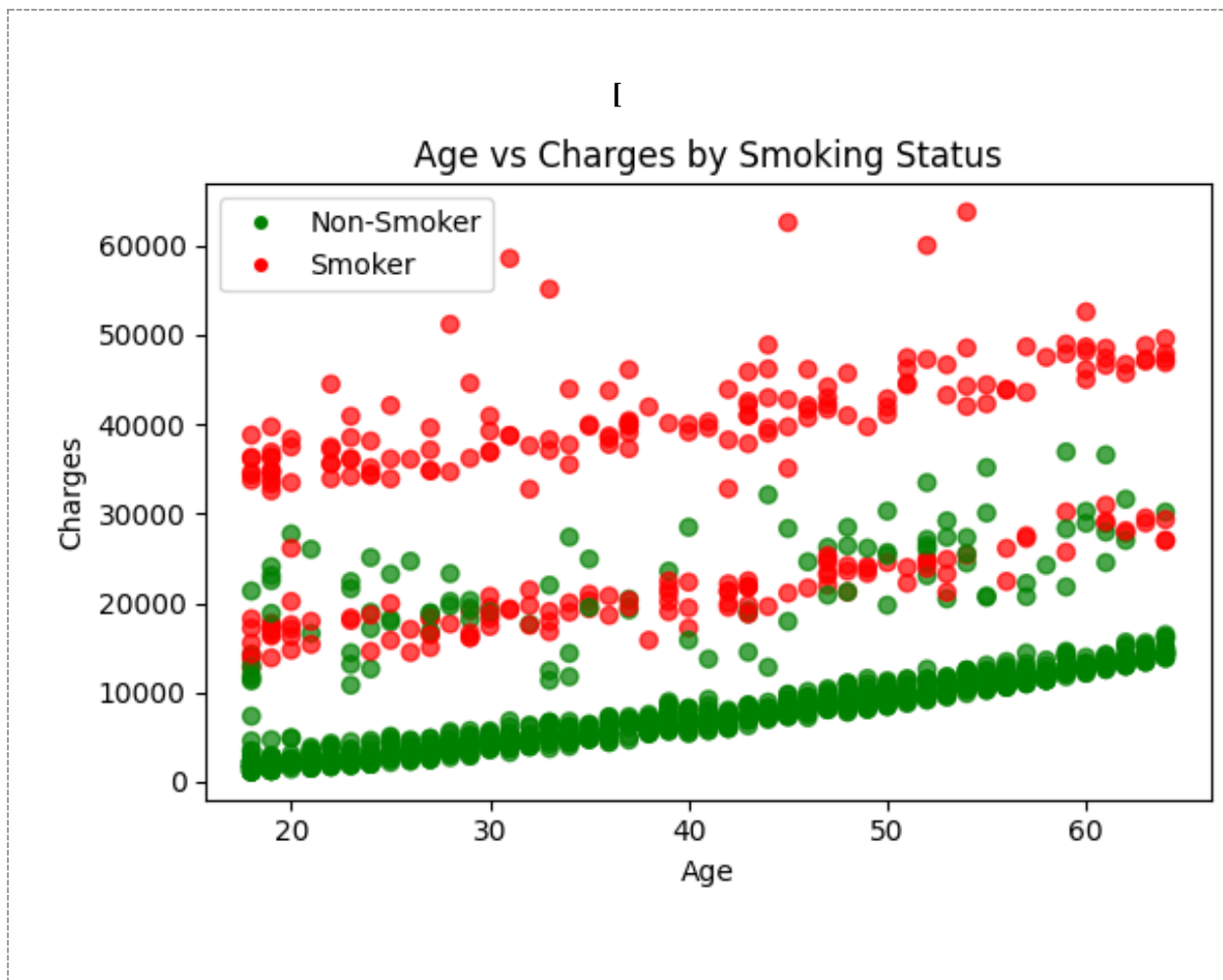


Figure 4

— Age vs. charges, coloured by smoker status, shows two distinct upward trends — smokers' costs rise faster with age than non-smokers'.

6. Model Selection & Optimization

Baseline Model: Since this project follows the **regression** track, **Ridge Regression** was selected as the baseline model. Ridge Regression is a regularized version of Linear Regression that reduces overfitting by penalizing large coefficients. Using the preprocessed dataset, the baseline model achieved a

test R^2 score of 0.51, providing a benchmark for evaluating more advanced models.

Second Model: A **Random Forest Regressor** was selected as the second model because it effectively captures complex, non-linear relationships between features such as age, BMI, smoking status, and insurance charges. It is also robust to outliers, handles mixed feature types well, and generally provides higher predictive accuracy on tabular datasets than linear models.

Optimization: Both models were optimized using **GridSearchCV** with **5-fold cross-validation**. For Ridge Regression, the hyperparameters explored included **alpha (1.0, 1.5, 2.0)**, **fit_intercept (True/False)**, and **max_iter**. For Random Forest, the search covered **n_estimators (100, 200, 300)**, **max_depth (None, 1, 2, 3)**, **criterion ("squared_error")**, and **max_samples (None)** to identify the most effective configuration.

Winning Configuration: The tuned **Random Forest Regressor** achieved the best performance with **100 estimators**, **maximum depth of 3**, **criterion = squared_error**, and **max_samples = None**, producing a **test R^2 score of approximately 0.86**. Hyperparameter tuning significantly improved prediction accuracy by controlling tree complexity, reducing overfitting, and enhancing the model's ability to generalize to unseen data.

(214 words)

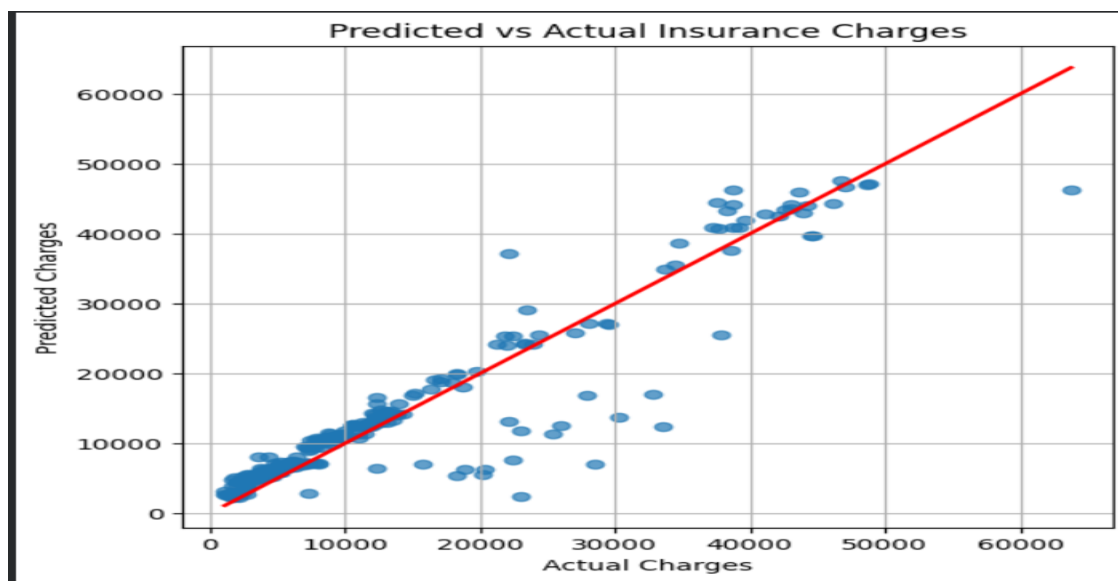
Model comparison:

Model	Preprocessing used	Test score
Ridge Regression (baseline, tuned)	Standard Scaler + one-hot encoding (sex, smoker, region)	$R^2 = 0.51$, RMSE $\approx$ \$8,681, MAE $\approx$ \$4,224
Random Forest — default settings	Standard Scaler + one-hot encoding (sex, smoker, region)	$R^2 = 0.86$, RMSE $\approx$ \$4,583, MAE $\approx$ \$2,542
Random Forest — after tuning (GridSearchCV)	Standard Scaler + one-hot encoding (sex, smoker, region)	$R^2 = 0.86$, RMSE $\approx$ \$4,598, MAE $\approx$ \$2,347

Table 1 — The tuned Random Forest was selected: it roughly halves both MAE and RMSE versus the Ridge baseline. Add the untuned Random Forest row above for a complete before/after picture.

7. Model Evaluation

Since this project addresses a **regression** problem, the model was evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and the **coefficient of determination (R^2)**. The tuned Random Forest Regressor achieved **MAE $\approx$ \$2,347**, **RMSE $\approx$ \$4,598**, and **$R^2 = 0.86$** , outperforming the Ridge Regression baseline. In practical terms, an MAE of approximately **\$2,347** means that the model's predictions differ from the actual annual insurance charges by about **\$2,347 on average**, which is acceptable given the wide range of insurance costs in the dataset. During model development, the Random Forest obtained a training R^2 of approximately **0.81**, while the test R^2 reached **0.86**. The small difference between these scores does not indicate significant overfitting, suggesting that the model generalizes well to unseen data and provides reliable predictions.



5 Figure — Predicted vs. actual charges for the tuned Random Forest cluster tightly along the diagonal for lower charges, with more spread at the high end.

8. Deployment (Demo)

The trained Random Forest model was saved using the `joblib.dump()` I do it on gradio function to enable future reuse without retraining. A simple Streamlit application was developed to provide an interactive prediction interface. Users enter age, sex, BMI, number of children, smoking status, and region, and the application processes these inputs through the trained model to estimate annual medical insurance charges. Figure 6 shows the application successfully generating a prediction for a sample user.

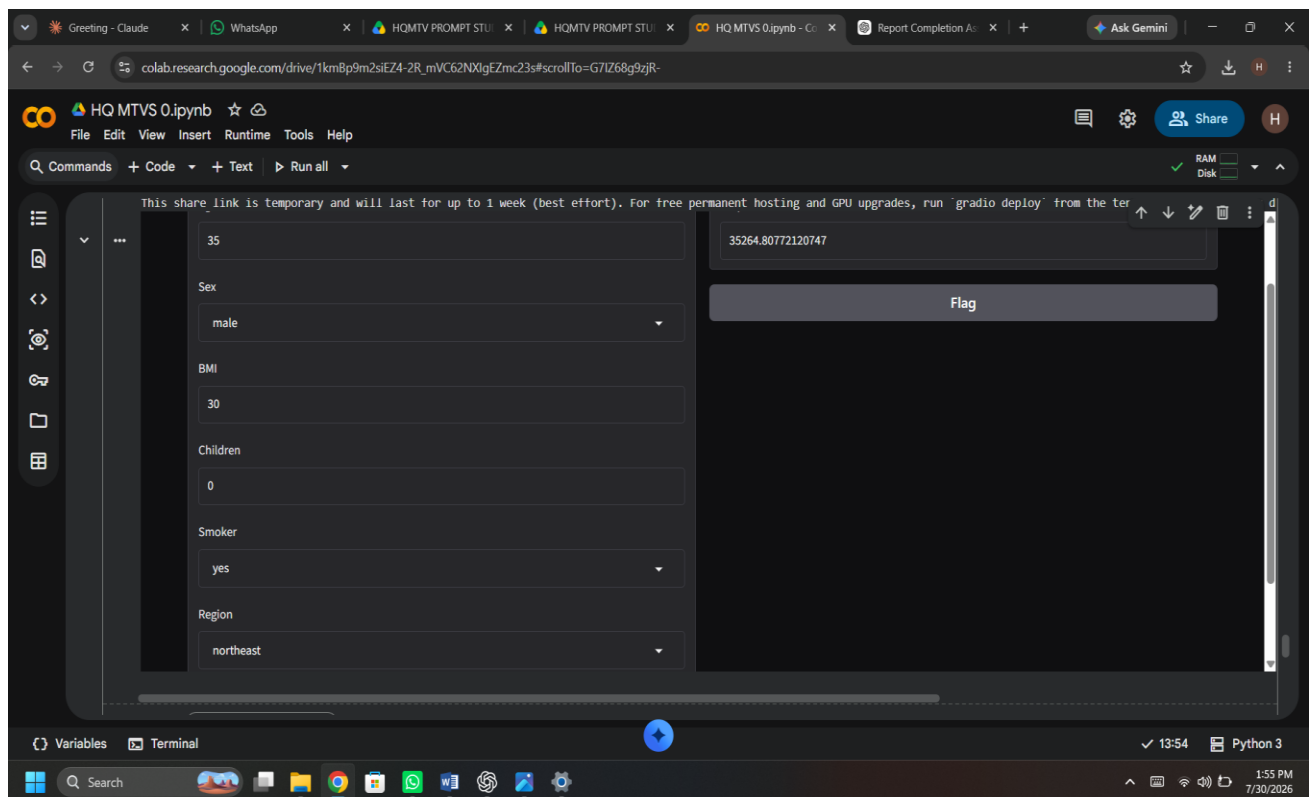


Figure 6. Screenshot of the deployed prediction application showing sample user inputs (Age = 35, Sex = Male, BMI = 30, Children = 0, Smoker = yes, Region = northeast) and the corresponding predicted annual medical insurance charges generated by the trained Random Forest model. The interface demonstrates how users can enter their information and instantly receive an estimated insurance cost.

9. Conclusion & Learnings

This project successfully developed a machine learning model to predict medical insurance charges using demographic and health-related features. After comparing multiple models, the tuned Random Forest Regressor achieved the best overall performance, with an R^2 score of approximately 0.86 and substantially lower prediction error than the Ridge Regression baseline. Exploratory Data Analysis showed that smoking status and age have the strongest influence on insurance costs. Through this project, we gained practical experience in data preprocessing, feature encoding, model selection, hyperparameter tuning, and model evaluation. Future work may include testing additional ensemble models and deploying the trained model as a web application for real-time predictions.

10. References & Links

- Colab notebook: [https://colab.research.google.com/drive/1kmBp9m2siEZ4-2R_mVC62NXIgEZmc23s]
- • Kaggle. *Medical Cost Personal Insurance Dataset*.
<https://www.kaggle.com/datasets/mirichoi0218/insurance>
- • Scikit-learn Documentation.
<https://scikit-learn.org/stable/>
- • Pandas Documentation.
<https://pandas.pydata.org/docs/>
- • Matplotlib Documentation.
<https://matplotlib.org/stable/>
- • NumPy Documentation.
<https://numpy.org/doc/>
- • YData Profiling Documentation.
<https://docs.profiling.ydata.ai/latest/>
- Joblib Documentation
<https://joblib.readthedocs.io/>

- **KPITB AI & ML Training Program:** Week 3 Lecture – *From Clean Data to a Deployed Model.*
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Thank you for your attention

